Gastroesophageal Reflux Disease (GERD)

All key points for the exam + drug summary table

1. Definition

- **GERD:** reflux of stomach contents into the esophagus causing **troublesome symptoms** and/or **complications**.
- **Heartburn** (burning behind the sternum) is the most common symptom.
- Many people have occasional heartburn; it becomes **GERD** when it is **frequent or troublesome** (e.g., ≥ 2 days/week) or causes damage.

Type	Description
Non-erosive reflux disease (NERD)	Symptoms but **no erosions** on endoscopy (most common, ~60–70%)
Erosive esophagitis (EE)	**Visible erosions** on endoscopy • Graded by the **Los Angeles (LA) classification A–D** (D = most severe)
Barrett's esophagus	Metaplasia of esophageal lining (squamous → **columnar**) • **Pre-cancerous** (esophageal adenocarcinoma)

2. Pathophysiology

- **Transient lower esophageal sphincter relaxations (TLESRs) — most common mechanism**.
- **Decreased LES pressure** (foods, drugs, smoking).
- **Hiatal hernia**.
- **Impaired esophageal clearance** (↓ saliva, poor motility).
- **Delayed gastric emptying**.
- **Increased intra-abdominal pressure** (**obesity**, **pregnancy**, tight clothing).
- **Acid**, pepsin, and bile cause the mucosal damage.

3. Clinical Presentation

Type	Symptoms
Typical (esophageal)	**Heartburn**, **regurgitation** (sour/bitter taste), hypersalivation (water brash), belching • Worse **after meals**, **lying down**, or **bending over**
Atypical (extraesophageal)	**Chronic cough**, **asthma** (worsening), **hoarseness/laryngitis**, sore throat, **non-cardiac chest pain**, **dental erosions**, recurrent pneumonia, sinusitis
Alarm symptoms (complicated GERD)	**Dysphagia** (difficulty swallowing), **odynophagia** (painful swallowing), **unintentional weight loss**, **GI bleeding** (hematemesis, melena), **anemia**, persistent vomiting, choking

Alarm symptoms → refer for endoscopy (not self-treatment). Also rule out **cardiac chest pain** first: chest pain with shortness of breath, sweating, or radiation to the arm/jaw is an **emergency**.

4. Risk Factors and Contributing Factors

Factors that decrease LES pressure

Category	Examples
Foods / drinks	**Fatty meals**, **chocolate**, **peppermint/spearmint**, **caffeine/coffee**, **alcohol**, garlic, onions, carbonated drinks
Lifestyle	**Smoking**, **obesity**, large meals, lying down after meals, tight clothing
Drugs (↓ LES pressure)	**Anticholinergics**, **calcium channel blockers** (dihydropyridines), **nitrates**, **theophylline**, **benzodiazepines**, **opioids**, barbiturates, **estrogen/progesterone**, **beta-2 agonists**, alpha-blockers, **TCAs**, dopamine

Drugs that directly irritate the esophagus (pill esophagitis)

- **NSAIDs and aspirin**
- **Bisphosphonates** (alendronate, risedronate, ibandronate)
- **Potassium chloride**
- **Iron** supplements
- **Tetracyclines** (especially **doxycycline**)
- Quinidine, zidovudine, some chemotherapy

Counseling to prevent pill esophagitis: take with a **full glass of water** and **stay upright for 30 minutes** (bisphosphonates: 30–60 min depending on the drug).

Other risk factors

Pregnancy, **hiatal hernia**, scleroderma, Zollinger-Ellison syndrome, gastroparesis, older age.

5. Diagnosis

Method	When / key points
Clinical diagnosis	**Typical symptoms (heartburn, regurgitation) → empiric PPI trial for 8 weeks** (diagnostic and therapeutic)
Upper endoscopy (EGD)	**Alarm symptoms** • **No response** to PPI twice daily • Screening for Barrett's in high-risk patients (e.g., men, > 50 years, white, obesity, smoking, chronic GERD, family history) • Grades esophagitis (LA A–D)
Ambulatory pH monitoring	Confirms GERD when endoscopy is normal (NERD) and before surgery • **Stop PPIs 7 days** before testing (off-therapy)
Esophageal manometry	**Before anti-reflux surgery** (rules out motility disorders such as achalasia)
Barium swallow	**Not** for diagnosing GERD (may help evaluate dysphagia/strictures)

6. Complications

Complication	Key points
Erosive esophagitis	Needs **8 weeks of PPI** to heal • LA grade C/D → **long-term maintenance PPI**
Esophageal stricture	Dysphagia • **Endoscopic dilation** + **long-term PPI**
Barrett's esophagus	Risk of **esophageal adenocarcinoma** • **Long-term (indefinite) PPI** • **Surveillance endoscopy** • Endoscopic eradication for dysplasia
Bleeding, ulceration	Anemia, melena, hematemesis
Extraesophageal	Asthma, chronic cough, laryngitis, dental erosion

7. Treatment Goals

- **Relieve symptoms**.
- **Heal** the esophagus (erosive esophagitis).
- **Prevent recurrence** and **complications**.
- Use the **lowest effective dose** for the **shortest duration**.

8. Lifestyle Modifications (for ALL patients)

Measure	Notes
Weight loss	If overweight/obese — **strongest evidence**
Elevate the head of the bed	By **6–8 inches** (blocks or wedge; extra pillows are not enough) — for **nighttime symptoms**
Avoid meals 2–3 hours before bedtime	For nighttime symptoms
Avoid trigger foods	Only foods that trigger symptoms **for that patient** (routine elimination of all triggers is not required)
Smoking cessation	↑ LES pressure, ↑ saliva
Limit alcohol	
Smaller meals, avoid tight clothing	
Review medications	Stop or replace drugs that ↓ LES pressure or irritate the esophagus if possible

9. Treatment Approach

Self-care (OTC) and when to refer

Situation	Management
Mild, infrequent heartburn (< 2 days/week)	Lifestyle + **antacids** (fast relief) or **H2RA** as needed (OTC) • H2RA can be taken **30–60 min before** a trigger meal to **prevent** heartburn
Frequent heartburn (≥ 2 days/week)	**OTC PPI once daily for 14 days** • May repeat a 14-day course **every 4 months** (max **3 courses per year**)
Refer to a physician	**Alarm symptoms** • Symptoms **> 3 months** • Symptoms persisting **after 14 days** of OTC PPI or needing more than 3 courses/year • Heartburn while taking OTC H2RA/PPI • **Atypical symptoms** (cough, wheezing, chest pain) • **Pregnancy**, children (OTC PPIs: age **≥ 18**; OTC H2RAs: **≥ 12**)

Prescription therapy (ACG guideline)

1. **PPI once daily for 8 weeks**, taken **30–60 minutes before the first meal** of the day (breakfast).
2. **Partial response** → check **adherence and timing** → increase to **PPI twice daily** (before breakfast and dinner) **or switch** to another PPI.
3. **After 8 weeks**, if symptoms resolve:
4. – **NERD:** try to **stop** the PPI, or use the **lowest effective dose** or **on-demand** PPI.
5. – **Erosive esophagitis LA C/D**, **Barrett's**, **stricture**: **long-term maintenance PPI**.
6. **No response to PPI twice daily** (refractory) → **endoscopy** and **pH monitoring**.
7. **Nighttime breakthrough symptoms** → may add **bedtime H2RA** (tolerance develops; use intermittently).
8. **Regurgitation-predominant**, refractory → consider **baclofen** or surgery.

Step-up vs step-down

Strategy	Description
Step-up	Start with lifestyle/antacids/H2RA, escalate to PPI if needed — for **mild, infrequent** symptoms
Step-down	Start with **PPI** (fast control and healing), then **reduce** to the lowest effective dose or H2RA/on-demand — for **frequent or severe** symptoms (**most common approach**)

Surgery and procedures

- **Laparoscopic fundoplication** (Nissen) • **Magnetic sphincter augmentation (LINX)** • Transoral incisionless fundoplication (TIF).
- For patients who **do not want lifelong medication**, have **regurgitation** despite PPI, large hiatal hernia, or PPI intolerance.
- Requires **pH monitoring and manometry** beforehand.

10. Antacids

- **Mechanism: neutralize** stomach acid.
- **Onset: fastest** (within 5 minutes) • **Duration:** short (**30–60 min** on an empty stomach; **up to 3 hours** if taken after meals).
- **Use: mild, infrequent** heartburn; **do not heal** erosive esophagitis.
- **Liquids** work faster than tablets (tablets must be chewed well).

Antacid	Key ADRs / notes
Aluminum hydroxide	**Constipation** • **Hypophosphatemia** • **Aluminum accumulation in CKD** (encephalopathy, osteomalacia)
Magnesium hydroxide	**Diarrhea** • **Hypermagnesemia in CKD** (avoid)
Aluminum + magnesium (e.g., Maalox, Mylanta)	Combination balances constipation and diarrhea
Calcium carbonate (Tums)	**Constipation**, gas • **Hypercalcemia**, **milk-alkali syndrome**, kidney stones • **Acid rebound**
Sodium bicarbonate	**High sodium load** (avoid in **heart failure, hypertension, edema**, low-sodium diet) • **Systemic alkalosis** • Gas, bloating • **Avoid in pregnancy**
Alginic acid (Gaviscon)	Forms a **foam barrier (raft)** on top of stomach contents • Good for **regurgitation** • Works best **upright**
Simethicone	Anti-gas only (no acid effect)

Antacid drug interactions (high-yield)

- **Chelation** (↓ absorption): **tetracyclines**, **fluoroquinolones**, **integrase inhibitors** (dolutegravir, bictegravir, raltegravir), **iron**, **levothyroxine**, bisphosphonates, mycophenolate.
- **↑ Gastric pH** → ↓ absorption of **ketoconazole**, **itraconazole**, **atazanavir**, **rilpivirine**, cefuroxime axetil, dasatinib and other pH-dependent drugs.
- **Separate** by at least **2 hours** (fluoroquinolones: **2 h before or 6 h after** antacids).

11. Histamine-2 Receptor Antagonists (H2RAs)

- **Mechanism: competitively block H2 receptors** on parietal cells → ↓ acid (especially **nocturnal** acid).
- **Onset:** 30–60 minutes • **Duration:** about **4–10 hours** (up to 12).
- **Use:** mild–moderate GERD, **on-demand**, **prevention** before trigger meals, **bedtime add-on** for nocturnal symptoms.
- **Less effective** than PPIs for healing erosive esophagitis.
- **Tachyphylaxis (tolerance)** develops within **2 weeks** of continuous use.
- **Ranitidine** was **withdrawn (2020)** because of **NDMA** (carcinogen) contamination • Nizatidine availability is limited.

Drug	OTC dose	Prescription GERD dose
Famotidine (Pepcid)	**10–20 mg** up to twice daily	**20 mg BID** (up to 6 weeks) • Erosive esophagitis: **20–40 mg BID** (up to 12 weeks)
Cimetidine (Tagamet)	**200 mg** up to twice daily	**400 mg BID** or **800 mg BID** (erosive)
Nizatidine (Axid)	75 mg up to twice daily	**150 mg BID**

Adverse effects

- Generally well tolerated: **headache**, dizziness, diarrhea or constipation.
- **CNS effects** (**confusion**, delirium, hallucinations) — in **elderly** and **renal impairment** → **reduce dose** (famotidine: ↓ dose if **CrCl < 60**).
- **Famotidine: QT prolongation** in renal impairment.
- Rare: thrombocytopenia, ↑ liver enzymes.
- **Cimetidine: antiandrogenic** effects (**gynecomastia**, **impotence**, galactorrhea) with high doses; **↑ serum creatinine** (blocks tubular secretion, not true kidney injury).

Drug interactions

- **Cimetidine** = **CYP inhibitor** (1A2, 2C9, 2C19, 2D6, 3A4) → ↑ **warfarin**, **theophylline**, **phenytoin**, **lidocaine**, **nifedipine**, TCAs, benzodiazepines → **famotidine is preferred** (no CYP inhibition).
- All H2RAs: **↑ pH** → ↓ absorption of **ketoconazole, itraconazole, atazanavir, rilpivirine**, dasatinib, iron, cefuroxime axetil.

12. Proton Pump Inhibitors (PPIs) — drugs of choice

Mechanism and pharmacology

- **Irreversibly inhibit the H^+/K^+-ATPase (proton pump)** — the **final step** of acid secretion → **most potent** acid suppression.
- **Prodrugs** activated in acid → they block only **active pumps** → take **30–60 minutes before a meal** (usually **breakfast**).
- **Delayed-release** (enteric-coated) to protect them from stomach acid.
- Full effect after **2–5 days** (not ideal for immediate on-demand relief).
- Metabolized by **CYP2C19** (and 3A4) → genetic variation: **poor metabolizers** have higher levels; **rapid metabolizers** may respond poorly.
- **Most effective** for **healing erosive esophagitis** and maintaining remission.

Doses

Drug	Standard dose (once daily)	Notes
Omeprazole (Prilosec)	**20 mg** daily (40 mg for EE in some regimens)	OTC 20 mg • Strong **CYP2C19 inhibitor**
Omeprazole + sodium bicarbonate (Zegerid)	20–40 mg daily	**Immediate-release** • Take on an **empty stomach 1 h before a meal** • **Sodium content**
Esomeprazole (Nexium)	**20–40 mg** daily	OTC 20 mg • IV available • CYP2C19 inhibitor
Lansoprazole (Prevacid)	**15–30 mg** daily (30 mg for EE)	OTC 15 mg • **ODT** (orally disintegrating) available
Dexlansoprazole (Dexilant)	**30 mg** (NERD) • **60 mg** (EE healing)	**Dual delayed-release** → **can be taken without regard to meals**
Pantoprazole (Protonix)	**40 mg** daily	**IV available** • **Fewest CYP interactions** • Preferred with **clopidogrel**
Rabeprazole (AcipHex)	**20 mg** daily	Less CYP2C19-dependent • Sprinkle capsules available

- **Twice-daily dosing:** before **breakfast and dinner** (for partial response or extraesophageal symptoms).

- **OTC PPIs** (omeprazole 20 mg, esomeprazole 20 mg, lansoprazole 15 mg): **once daily × 14 days**, ages ≥ 18.
- **No renal dose adjustment** • Consider dose reduction in **severe hepatic impairment**.

Administration

- Swallow **whole**; **do not crush or chew** delayed-release products.
- Capsules (**omeprazole, esomeprazole, lansoprazole, dexlansoprazole, rabeprazole sprinkle**) can be **opened and sprinkled on applesauce** for patients who cannot swallow.
- **Nasogastric (NG) tube:** use **suspension packets** (esomeprazole, omeprazole, pantoprazole granules) or **lansoprazole ODT** dispersed in water.
- **IV PPIs** (pantoprazole, esomeprazole) for patients who cannot take oral medications (and GI bleeding).

Adverse effects

Type	Adverse effects
Short-term (common)	**Headache**, **diarrhea**, nausea, abdominal pain, constipation, flatulence
Long-term (associations)	**C. difficile infection** • **Pneumonia** (community-acquired) • **Bone fractures** (hip, spine; ↓ calcium absorption) • **Hypomagnesemia** (can cause arrhythmias, seizures; check Mg if long-term or with diuretics/digoxin) • **Vitamin B12 deficiency** • **Iron deficiency** • **Acute interstitial nephritis (AIN)** • Chronic kidney disease (association) • Fundic gland polyps • Cutaneous and systemic **lupus erythematosus** • Possible **SIBO**
On stopping	**Rebound acid hypersecretion → taper** the PPI (e.g., reduce dose or every other day over 2–4 weeks) with antacids/H2RA as needed

Calcium supplement + PPI: use **calcium citrate** (absorbed without acid) instead of calcium carbonate.

Drug interactions (high-yield)

Interacting drug	Interaction
Clopidogrel	**Omeprazole and esomeprazole** inhibit **CYP2C19** → ↓ activation of clopidogrel → ↓ antiplatelet effect → **avoid**; use **pantoprazole** (or rabeprazole/H2RA famotidine)
pH-dependent drugs	↓ Absorption: **ketoconazole**, **itraconazole**, posaconazole suspension, **atazanavir**, **rilpivirine (contraindicated with PPIs)**, nelfinavir, **ledipasvir/velpatasvir**, **dasatinib, erlotinib** and other TKIs, **iron**, **calcium carbonate**, cefuroxime axetil, **mycophenolate mofetil**, levothyroxine
Methotrexate (high dose)	↑ Methotrexate levels → toxicity
Omeprazole (CYP2C19 inhibition)	↑ **citalopram/escitalopram** (QT), ↑ **diazepam**, ↑ **phenytoin**, ↑ warfarin (minor), ↑ **tacrolimus**
CYP inducers	**St. John's wort**, **rifampin** ↓ PPI levels
Digoxin	↑ Digoxin absorption (minor) • hypomagnesemia ↑ digoxin toxicity
Diuretics	↑ Risk of **hypomagnesemia**

Deprescribing PPIs

- Reassess the need for PPI regularly (many patients do not need long-term therapy).
- **Do NOT deprescribe** in: **Barrett's esophagus**, **severe (LA C/D) erosive esophagitis**, **history of GI bleeding**, chronic NSAID/aspirin users at high GI risk.
- **Method:** reduce dose by 50% or take every other day for 2–4 weeks, then stop; or switch to **on-demand** use.
- Use antacids/H2RA for rebound symptoms.

13. Potassium-Competitive Acid Blocker (P-CAB)

Point	Details
Drug	**Vonoprazan** (Voquezna)
Mechanism	**Reversibly** blocks the proton pump by competing with potassium • **Not a prodrug**, **acid-stable**
Advantages over PPIs	**Faster onset** and **more potent, longer** acid suppression • Can be taken **with or without food** • Not affected by CYP2C19 polymorphism
Doses	**Erosive esophagitis healing: 20 mg daily × 8 weeks** • **Maintenance: 10 mg daily** (up to 6 months) • **NERD heartburn: 10 mg daily × 4 weeks** • Also used in **H. pylori** regimens
ADRs	Similar to PPIs (diarrhea, abdominal pain, headache; long-term risks similar)
DDIs	**CYP3A4 substrate** → avoid strong/moderate **CYP3A4 inducers** • ↓ absorption of pH-dependent drugs • **Contraindicated with rilpivirine**

14. Other Agents

Drug	Key points
Metoclopramide (Reglan)	**Prokinetic** (dopamine antagonist) • **Not recommended** for GERD unless **gastroparesis** is present • Dose: 10 mg up to QID, 30 min before meals and at bedtime • **Boxed warning: tardive dyskinesia** → **avoid use > 12 weeks** • EPS, drowsiness, depression, hyperprolactinemia • **Reduce dose in renal impairment** • Avoid in Parkinson's disease
Baclofen	**GABA-B agonist** → ↓ **TLESRs** • For **refractory regurgitation** • 5–20 mg TID • **CNS effects**: drowsiness, dizziness, confusion • **Avoid in renal impairment** • Do not stop abruptly (withdrawal, seizures)
Sucralfate (Carafate)	Mucosal protectant (coats damaged mucosa) • **Limited role in GERD** (mainly pregnancy) • **1 g QID** on an empty stomach • **Constipation** • Contains **aluminum** (avoid long-term in CKD) • **Binds many drugs** (quinolones, levothyroxine, phenytoin, warfarin, digoxin) → **separate by 2 hours**
Bethanechol	Cholinergic agonist • **Not recommended** (ADRs, little benefit)

15. Special Populations

Population	Key points
Pregnancy	Very common • **Step-up** approach: **lifestyle** → **antacids** (calcium- or magnesium-based; **avoid sodium bicarbonate** and **magnesium trisilicate**) or **sucralfate** → **H2RA (famotidine)** → **PPI** if needed (considered safe) • Avoid metoclopramide long-term
Infants	Regurgitation ("**happy spitters**") is normal and resolves by ~12–18 months • **No PPIs/H2RAs** for uncomplicated regurgitation • **Thickened feeds**, smaller frequent feeds, trial of cow's-milk-protein-free formula
Children / adolescents	PPIs approved from age 1 for erosive esophagitis • OTC PPIs only ≥ 18 years
Elderly	Higher risk of PPI long-term complications (fractures, C. diff, B12) • **Avoid cimetidine** • Adjust H2RAs for renal function • **Beers criteria:** avoid PPIs > 8 weeks unless clear indication • Avoid metoclopramide
Extraesophageal GERD	If typical symptoms also present → **PPI twice daily for 8–12 weeks** • Without typical symptoms → test for reflux before long-term PPI
Barrett's esophagus	**Indefinite PPI** (at least once daily) + **surveillance endoscopy**

16. GERD vs Peptic Ulcer Disease and H. pylori

- **H. pylori** is **not** a cause of GERD, and eradication **does not improve** GERD.
- Do **not** confuse GERD dosing (once daily before breakfast, 8 weeks) with **H. pylori** regimens (PPI **twice daily** with antibiotics for 14 days).
- **NSAID users** at high GI risk: PPI for **ulcer prevention**.

Exam Traps and Frequently Tested Points

1. **Most common symptom:** heartburn • **Most common mechanism: transient LES relaxations**.
2. **Alarm symptoms** (dysphagia, odynophagia, weight loss, GI bleeding, anemia) → **endoscopy**, not self-care.
3. **Typical symptoms** → **empiric PPI trial for 8 weeks** (no endoscopy needed first).
4. **Weight loss** has the strongest lifestyle evidence • **Elevate head of bed 6–8 inches** for nighttime symptoms • **No meals 2–3 h before bed**.
5. **Drugs that ↓ LES pressure:** anticholinergics, CCBs, nitrates, theophylline, benzodiazepines, opioids, estrogen/progesterone.
6. **Pill esophagitis:** NSAIDs, bisphosphonates, KCl, iron, doxycycline → full glass of water, stay upright 30 min.
7. **Antacids:** fastest onset, shortest duration, **no healing**.
8. **Aluminum → constipation • Magnesium → diarrhea** • both accumulate in **CKD**.
9. **Sodium bicarbonate** → avoid in HF, HTN, pregnancy • **Calcium carbonate** → milk-alkali syndrome.
10. **Alginic acid (Gaviscon)** → raft, good for **regurgitation**.
11. **Antacids/H2RAs/PPIs + quinolones, tetracyclines, INSTIs, azoles, atazanavir** → ↓ absorption.
12. **H2RAs:** good for **prevention** (30–60 min before meals) and **nocturnal** acid • **tachyphylaxis** in 2 weeks.
13. **Cimetidine** → CYP inhibitor (warfarin, theophylline, phenytoin) + **gynecomastia** → prefer **famotidine**.
14. **H2RAs in elderly/renal impairment** → **confusion** → reduce dose.
15. **Ranitidine withdrawn** (NDMA).
16. **PPIs = most effective**; take **30–60 min before breakfast** (prodrugs need active pumps).
17. **Dexlansoprazole** → can be taken **without regard to meals**; **Zegerid** → empty stomach 1 h before meal.
18. **PPI therapy: once daily × 8 weeks** → partial response → **twice daily** or switch PPI.
19. **OTC PPI: 14 days**, up to **3 courses/year** (every 4 months); ≥ 18 years.
20. **Long-term maintenance PPI: LA C/D esophagitis**, **Barrett's**, strictures.
21. **Long-term PPI ADRs: C. diff**, pneumonia, **fractures**, **hypomagnesemia**, **B12 deficiency**, iron deficiency, AIN.
22. **PPI + calcium:** use **calcium citrate**.
23. **Clopidogrel + omeprazole/esomeprazole** → avoid; use **pantoprazole**.
24. **Rilpivirine + PPI** → **contraindicated**.
25. **Stopping PPIs** → **rebound acid hypersecretion** → **taper**.
26. **No renal dose adjustment** for PPIs; **H2RAs need renal adjustment**.
27. **Vonoprazan** (P-CAB): reversible, not a prodrug, with or without food, 20 mg × 8 weeks for EE.
28. **Metoclopramide:** not for GERD alone; **boxed warning tardive dyskinesia**, max 12 weeks.
29. **Baclofen** → ↓ TLESRs, for refractory regurgitation; CNS side effects.
30. **Sucralfate** → binds drugs (separate 2 h); option in pregnancy.
31. **Pregnancy:** lifestyle → antacids/sucralfate → famotidine → PPI; **avoid sodium bicarbonate and magnesium trisilicate**.
32. **Infants who spit up** ("happy spitters") → **no acid suppression**.
33. **pH monitoring** → stop PPI **7 days** before • **Manometry** before surgery.
34. **Barrett's esophagus** → risk of adenocarcinoma → indefinite PPI + surveillance.
35. **Rule out cardiac causes** of chest pain before attributing it to GERD.

Note: Some doses and cut-offs differ slightly between references. **Use the numbers in the reference you are studying for the exam.**

Drug Summary Table: Doses, Key ADRs and DDIs

Drug	Dose	Key ADRs	Key DDIs / special notes
Aluminum hydroxide	PRN (e.g., 300–600 mg) after meals and at bedtime	**Constipation**, **hypophosphatemia**, Al accumulation in CKD	**Chelates** quinolones, tetracyclines, INSTIs, iron, levothyroxine → separate ≥ 2 h
Magnesium hydroxide	PRN	**Diarrhea**, **hypermagnesemia** (CKD)	Same chelation • Avoid in CKD
Calcium carbonate (Tums)	500–1,000 mg PRN (max ~7 g/day calcium carbonate OTC)	Constipation, gas, **hypercalcemia**, **milk-alkali syndrome**, acid rebound	Same chelation • ↓ absorption with PPIs if used as a supplement
Sodium bicarbonate	PRN, short-term only	**Sodium load**, **metabolic alkalosis**, gas	Avoid in **HF, HTN, edema, pregnancy**
Alginic acid (Gaviscon)	PRN after meals and at bedtime	GI effects (minimal)	Forms raft; best for **regurgitation**; take upright
Famotidine (Pepcid)	OTC 10–20 mg up to BID • GERD 20 mg BID • EE 20–40 mg BID	Headache, dizziness • **Confusion** (elderly, renal) • **QT prolongation** (renal impairment)	**Reduce dose if CrCl < 60** • ↓ absorption of azoles, atazanavir, rilpivirine (separate) • **No CYP inhibition**
Cimetidine (Tagamet)	OTC 200 mg up to BID • 400–800 mg BID	**Gynecomastia**, **impotence**, confusion, ↑ SCr (false)	**CYP inhibitor** → ↑ **warfarin, theophylline, phenytoin, lidocaine**, TCAs • Renal dose adjustment
Nizatidine (Axid)	150 mg BID (OTC 75 mg)	Headache, confusion (renal)	Renal dose adjustment • pH-dependent DDIs
Omeprazole (Prilosec)	20 mg daily before breakfast (OTC 20 mg × 14 days)	Headache, diarrhea • Long-term: **C. diff, fractures, hypomagnesemia, B12 deficiency**, pneumonia, AIN	**CYP2C19 inhibitor** → ↓ **clopidogrel** (avoid) • ↑ citalopram, diazepam, phenytoin, tacrolimus, MTX • ↓ azoles, atazanavir, **rilpivirine (CI)**, iron, TKIs, MMF
Omeprazole/sodium bicarbonate (Zegerid)	20–40 mg daily, empty stomach 1 h before meal	Same as PPIs + **sodium load**	Same as omeprazole • Caution in sodium-restricted patients
Esomeprazole (Nexium)	20–40 mg daily (OTC 20 mg × 14 days)	Same as PPIs	Same as omeprazole (**avoid with clopidogrel**) • IV available
Lansoprazole (Prevacid)	15–30 mg daily (OTC 15 mg × 14 days)	Same as PPIs	pH-dependent DDIs • ODT available (NG tube)
Dexlansoprazole (Dexilant)	30 mg (NERD), 60 mg (EE) daily	Same as PPIs	**Dual delayed-release; no regard to meals**
Pantoprazole (Protonix)	40 mg daily (PO/IV)	Same as PPIs	**Fewest CYP interactions • Preferred with clopidogrel**
Rabeprazole (AcipHex)	20 mg daily	Same as PPIs	Less CYP2C19 dependence • pH-dependent DDIs
Vonoprazan (Voquezna)	EE: 20 mg daily × 8 wk • Maintenance 10 mg daily • NERD 10 mg daily × 4 wk	Diarrhea, abdominal pain, headache (long-term risks like PPIs)	**CYP3A4 substrate** (avoid strong/moderate inducers) • **Contraindicated with rilpivirine** • With or without food
Metoclopramide (Reglan)	10 mg up to QID, 30 min before meals and HS (max 12 weeks)	**Boxed: tardive dyskinesia** • EPS, drowsiness, depression, hyperprolactinemia	Renal dose adjustment • Avoid with antipsychotics, in Parkinson's, seizures • ↑ absorption of some drugs

Drug	Dose	Key ADRs	Key DDIs / special notes
Baclofen	5–20 mg TID	**Drowsiness**, dizziness, confusion	↑ CNS depression with alcohol/sedatives • **Avoid in renal impairment** • Taper (withdrawal)
Sucralfate (Carafate)	1 g QID (1 h before meals and HS)	**Constipation** • aluminum accumulation (CKD) • hyperglycemia (suspension)	**Binds** quinolones, levothyroxine, phenytoin, warfarin, digoxin, ketoconazole → **separate by 2 h**

Good luck on your exam!